

Self-Supervised Learning (SSL)

Introduction

Self-Supervised Learning (SSL) is a machine learning paradigm where models learn representations of data without requiring labeled examples. Instead of relying on human-annotated labels, SSL uses the inherent structure of the data to create supervisory signals. These learned representations can then be transferred to downstream tasks, such as classification, detection, or generation.

Key Concepts

1. Pretext Tasks

A pretext task is a surrogate task designed to train a model in a self-supervised manner. The goal is to learn meaningful representations of the data without requiring labeled data. Examples include:

- Image Rotation Prediction: Predict the rotation angle applied to an image.
- Masked Language Modeling: Predict masked tokens in a sentence.
- Contrastive Predictive Coding: Predict future audio frames from past frames.

2. Loss Functions

SSL uses loss functions to enforce consistency or similarity between different views or representations of the same input. Common loss functions include:

- Contrastive Loss: Encourages similar representations for positive pairs and dissimilar representations for negative pairs.
- Mean Squared Error (MSE) Loss: Minimizes the difference between predicted and target representations.
- Masked Language Modeling (MLM) Loss: Predicts masked tokens in the input text.

3. Augmentation

Random augmentation is a key technique in SSL that involves applying random transformations to input data to create different "views" of the same data. These views are used to train the model to learn robust and generalizable features. Examples include:

- Image Augmentation: Cropping, rotation, flipping, color jittering.
- Text Augmentation: Token masking, token swapping, synonym replacement.
- Audio Augmentation: Time stretching, pitch shifting, background noise.

SSL Techniques

1. Contrastive Learning

Contrastive learning is a popular SSL technique that trains models to distinguish between similar and dissimilar pairs of data. The model learns to produce similar representations for augmented views of the same input (positive pairs) and dissimilar representations for views of different inputs (negative pairs).

Example: SimCLR

- Pretext Task: Instance discrimination. - Loss Function: Contrastive loss. - Augmentation: Random cropping, color jittering, Gaussian blur.

2. Predictive Coding

Predictive coding involves predicting future data points from past data points. The model learns to capture the underlying structure of the data by predicting missing or future information.

Example: Contrastive Predictive Coding (CPC)

- Pretext Task: Predict future audio frames from past frames. - Loss Function: Contrastive loss. - Augmentation: Time masking, pitch shifting.

3. Reconstruction

Reconstruction-based SSL techniques train models to reconstruct the original input from a compressed or corrupted version. The model learns meaningful representations by minimizing the reconstruction error.

Example: Autoencoders

- Pretext Task: Reconstruct the original input from a compressed representation. - Loss Function: Mean squared error (MSE) loss. - Augmentation: Noise addition, masking.

Loss Functions in SSL

1. Contrastive Loss

- Purpose: Encourages similar representations for positive pairs and dissimilar representations for negative pairs. - Formula:

$$\mathcal{L} = -\log \frac{\exp(\text{sim}(\mathbf{h}_1, \mathbf{h}_2)/\tau)}{\sum_{k=1}^N \exp(\text{sim}(\mathbf{h}_1, \mathbf{h}_k)/\tau)}$$

where: - $\mathbf{h}_1$ and $\mathbf{h}_2$ are representations of two augmented views of the same input. - $\text{sim}(\cdot, \cdot)$ is a similarity function (e.g., cosine similarity). - τ is a temperature parameter. - N is the number of negative samples.

2. Mean Squared Error (MSE) Loss

- Purpose: Minimizes the difference between predicted and target representations. - Formula:

$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^N \|\mathbf{h}_1^{(i)} - \mathbf{h}_2^{(i)}\|^2$$

where $\mathbf{h}_1^{(i)}$ and $\mathbf{h}_2^{(i)}$ are representations of two augmented views of the same input.

3. Masked Language Modeling (MLM) Loss

- Purpose: Predicts masked tokens in the input text. - Formula:

$$\mathcal{L} = - \sum_{t \in \text{masked}} \log P(y_t \mid y_1, y_2, \dots, y_{t-1})$$

where: - y_t is the true token at position t . - $P(y_t \mid y_1, y_2, \dots, y_{t-1})$ is the predicted probability of the true token.

Pretext Tasks in SSL

1. Image Rotation Prediction

- Task: Predict the rotation angle applied to an image (e.g., 0°, 90°, 180°, 270°). - Example: Used in RotNet.

2. Masked Language Modeling (MLM)

- Task: Predict masked tokens in a sentence. - Example: Used in BERT.

3. Contrastive Predictive Coding (CPC)

- Task: Predict future audio frames from past frames. - Example: Used in CPC.

Applications of SSL

1. Computer Vision

- Image Classification: Pretrain models on large image datasets (e.g., ImageNet) and fine-tune on specific tasks. - Object Detection: Use SSL to learn robust features for detecting objects in images.

2. Natural Language Processing (NLP)

- Text Classification: Pretrain language models (e.g., BERT) on large text corpora and fine-tune on downstream tasks like sentiment analysis. - Machine Translation: Use SSL to learn cross-lingual representations for translation tasks.

3. Speech Recognition

- Speech-to-Text: Pretrain models on large audio datasets and fine-tune for speech recognition tasks. - Speaker Identification: Use SSL to learn speaker-specific features.

Key Takeaways

- SSL learns representations of data without requiring labeled examples. - Pretext tasks and loss functions are used to create supervisory signals. - Random augmentation is a key technique for creating diverse views of the data. - SSL has applications in computer vision, NLP, and speech recognition.

1 More Details on Loss Functions for Self-Supervised Learning in Text, Images, and Sound

Self-Supervised Learning (SSL) utilizes different **loss functions** depending on the type of data being processed:

- **Text (NLP Models like BERT, GPT)** → Cross-Entropy Loss for Masked Language Modeling
- **Images (Vision Models like SimCLR, MAE)** → Contrastive Loss or Reconstruction Loss
- **Sound (Speech Models like Wav2Vec 2.0)** → Contrastive Loss (InfoNCE)

1.1 Loss Function for Text (Masked Language Models - BERT, GPT)

Task: Predict missing words (Masked Language Modeling - MLM)

- **Example:** "The ___ is blue." → Model predicts the masked word.
- The loss function ensures that the predicted token distribution is close to the actual token.

Loss Function: Cross-Entropy Loss

$$\mathcal{L} = - \sum_{i \in M} \log p_{\theta}(y_i | x_{\text{masked}})$$

where:

- M is the set of masked positions.
- y_i is the ground-truth token at position i .
- $p_{\theta}(y_i | x_{\text{masked}})$ is the predicted probability of token y_i .

Used in: BERT, T5, ELECTRA, GPT (for next-word prediction instead of MLM).

1.2 Loss Function for Images (Contrastive Learning & Masked Autoencoders)

1.2.1 Contrastive Learning (SimCLR, MoCo)

- **Task:** Given two augmentations of the same image, maximize similarity.
- **Loss Function:** NT-Xent (Normalized Temperature-scaled Cross-Entropy Loss)

$$\mathcal{L} = - \log \frac{\exp(\text{sim}(z_i, z_j)/\tau)}{\sum_{k=1}^N \exp(\text{sim}(z_i, z_k)/\tau)}$$

where:

- z_i, z_j are embeddings of two augmented versions of the same image.
- $\text{sim}(z_i, z_j)$ is cosine similarity.
- τ is the temperature parameter.

Used in: SimCLR, MoCo, BYOL for visual representations.

1.2.2 Masked Autoencoder (MAE)

- **Task:** Randomly mask image patches and reconstruct them.
- **Loss Function:** Mean Squared Error (MSE)

$$\mathcal{L} = \frac{1}{n} \sum_{i=1}^n ||x_i - \hat{x}_i||^2$$

where:

- x_i is the true image patch.
- $\hat{x}_i$ is the reconstructed patch.

Used in: MAE (Masked Autoencoders for Vision).

1.3 Loss Function for Sound (Contrastive Speech Learning - Wav2Vec 2.0)

Task: Learn speech representations by distinguishing correct and incorrect future time steps.

- Instead of predicting words, the model learns useful speech features.

Loss Function: InfoNCE (Contrastive Loss for Audio)

$$\mathcal{L} = -\log \frac{\exp(\text{sim}(q, k^+)/\tau)}{\sum_{k \in K} \exp(\text{sim}(q, k)/\tau)}$$

where:

- q is the query (current speech embedding).
- k^+ is the **correct** future speech embedding.
- k are negative distractor embeddings.
- τ is the temperature parameter.

Used in: Wav2Vec 2.0, CPC (Contrastive Predictive Coding).

1.4 Final Summary

Data Type	SSL Task	Loss Function	Example Models
Text (NLP)	Masked Language Modeling	Cross-Entropy Loss	BERT, GPT
Image (Vision)	Contrastive Learning	NT-Xent Loss	SimCLR, MoCo, BYOL
Image (Vision)	Masked Image Prediction	MSE Loss	MAE
Audio (Speech)	Contrastive Speech Learning	InfoNCE Loss	Wav2Vec 2.0

Table 1: Summary of SSL Loss Functions for Different Modalities

Key Takeaway: Different SSL tasks use different loss functions, but they all aim to **learn useful representations from unlabeled data**.

2 Random Augmentations for Images in Self-Supervised Learning

In self-supervised learning for images, random **augmentations** are used to create different **views** of the same image. The goal is to make the model **learn invariant features**—features that remain meaningful despite transformations like cropping, rotation, or color changes.

These augmentations are essential in **contrastive learning frameworks** like **SimCLR**, **MoCo**, and **BYOL**, where the model learns by pulling similar views (positive pairs) closer and pushing different images (negative pairs) apart.

2.1 Geometric Transformations

These augmentations alter the spatial properties of the image.

Types:

- **Random Cropping & Resizing** → Selects a portion of the image and resizes it back.
- **Random Flipping** → Horizontally flips the image with a probability.
- **Random Rotation** → Rotates the image by a random degree (e.g., between -30° to 30°).
- **Random Affine Transformations** → Applies translation, scaling, shearing, or rotation.

Example: If you take a cat image and crop different parts of it, the model should still recognize it as a cat.

2.2 Photometric Transformations

These augmentations modify the **pixel values** of the image, affecting brightness, color, and contrast.

Types:

- **Color Jittering** → Randomly changes brightness, contrast, saturation, and hue.
- **Grayscale Conversion** → Converts the image to grayscale with a probability.
- **Gaussian Blur** → Applies a blur effect with a small kernel size to simulate out-of-focus effects.
- **Solarization** → Inverts pixel values above a threshold, creating a negative effect.

Example: The same cat image can have different brightness levels but should still be recognized as a cat.

2.3 Noise-Based Transformations

These augmentations introduce random distortions or noise to make the model robust to real-world variations.

Types:

- **Gaussian Noise** → Adds random pixel noise sampled from a Gaussian distribution.
- **Salt-and-Pepper Noise** → Randomly turns some pixels completely white or black.

Example: If an image is slightly noisy (e.g., from a bad camera), the model should still extract useful information.

2.4 Occlusion-Based Transformations

These augmentations block part of the image, forcing the model to learn from partial information.

Types:

- **Random Erasing** → Replaces a random region of the image with black pixels or noise.
- **Cutout** → Removes a fixed-size square patch from the image.
- **Mixup** → Combines two images by taking a weighted sum of their pixel values.
- **CutMix** → Cuts a region from one image and pastes it onto another.

Example: Even if part of the cat's face is covered, the model should still recognize it as a cat.

2.5 How These Augmentations Are Used in Self-Supervised Learning

- **SimCLR:** Uses a combination of *random cropping*, *flipping*, *color jittering*, and *Gaussian blur* to create different views of the same image.
- **MoCo:** Uses a *momentum encoder* and similar augmentations as SimCLR.
- **BYOL:** Uses *two networks* (online & target) and learns without explicit negative samples, still relying on strong augmentations.
- **MAE (Masked Autoencoders):** Uses *random masking* instead of contrastive learning to reconstruct missing image patches.

Key Takeaway: Random augmentations **force the model to learn meaningful representations** that are invariant to different transformations. These learned features **generalize well** for downstream tasks like object detection and classification.

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- **Masked Language Modeling**: Predict masked tokens in a sentence.
- **Contrastive Predictive Coding**: Predict future audio frames from past frames.

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Random augmentation is a key technique in SSL that involves applying random transformations to input data to create different "views" of the same data. These views are used to train the model to learn robust and generalizable features. Examples include:

- **Image Augmentation**: Cropping, rotation, flipping, color jittering.
- **Text Augmentation**: Token masking, token swapping, synonym replacement.
- **Audio Augmentation**: Time stretching, pitch shifting, background noise.

SSL Techniques

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Example: SimCLR

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Example: Contrastive Predictive Coding (CPC)

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3. Reconstruction

Reconstruction-based SSL techniques train models to reconstruct the original input from a compressed or corrupted version. The model learns meaningful representations by minimizing the reconstruction error.

Example: Autoencoders

- **Pretext Task**: Reconstruct the original input from a compressed representation. - **Loss Function**: Mean squared error (MSE) loss. - **Augmentation**: Noise addition, masking.

Loss Functions in SSL

1. Contrastive Loss

- **Purpose**: Encourages similar representations for positive pairs and dissimilar representations for negative pairs. - **Formula**:

$$\mathcal{L} = -\log \frac{\exp(\text{sim}(\mathbf{h}_1, \mathbf{h}_2)/\tau)}{\sum_{k=1}^N \exp(\text{sim}(\mathbf{h}_1, \mathbf{h}_k)/\tau)}$$

where: - $\mathbf{h}_1$ and $\mathbf{h}_2$ are representations of two augmented views of the same input. - $\text{sim}(\cdot, \cdot)$ is a similarity function (e.g., cosine similarity). - τ is a temperature parameter. - N is the number of negative samples.

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- **Purpose**: Minimizes the difference between predicted and target representations. - **Formula**:

$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^N \|\mathbf{h}_1^{(i)} - \mathbf{h}_2^{(i)}\|^2$$

where $\mathbf{h}_1^{(i)}$ and $\mathbf{h}_2^{(i)}$ are representations of two augmented views of the same input.

3. Masked Language Modeling (MLM) Loss

- **Purpose**: Predicts masked tokens in the input text. - **Formula**:

$$\mathcal{L} = - \sum_{t \in \text{masked}} \log P(y_t \mid y_1, y_2, \dots, y_{t-1})$$

where: - y_t is the true token at position t . - $P(y_t \mid y_1, y_2, \dots, y_{t-1})$ is the predicted probability of the true token.

Pretext Tasks in SSL

1. Image Rotation Prediction

- **Task**: Predict the rotation angle applied to an image (e.g., 0°, 90°, 180°, 270°). - **Example**: Used in **RotNet**.

2. Masked Language Modeling (MLM)

- **Task**: Predict masked tokens in a sentence. - **Example**: Used in **BERT**.

3. Contrastive Predictive Coding (CPC)

- **Task**: Predict future audio frames from past frames.
- **Example**: Used in **CPC**.

Applications of SSL

1. Computer Vision

- **Image Classification**: Pretrain models on large image datasets (e.g., ImageNet) and fine-tune on specific tasks.
- **Object Detection**: Use SSL to learn robust features for detecting objects in images.

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- **Text Classification**: Pretrain language models (e.g., BERT) on large text corpora and fine-tune on downstream tasks like sentiment analysis.
- **Machine Translation**: Use SSL to learn cross-lingual representations for translation tasks.

3. Speech Recognition

- **Speech-to-Text**: Pretrain models on large audio datasets and fine-tune for speech recognition tasks.
- **Speaker Identification**: Use SSL to learn speaker-specific features.

Key Takeaways

- SSL learns representations of data without requiring labeled examples.
- Pretext tasks and loss functions are used to create supervisory signals.
- Random augmentation is a key technique for creating diverse views of the data.
- SSL has applications in computer vision, NLP, and speech recognition.

More Details on Loss Functions for Self-Supervised Learning

Text (NLP Models like BERT, GPT)

- **Task**: Predict missing words (Masked Language Modeling - MLM).
- **Loss Function**: Cross-Entropy Loss.

$$\mathcal{L} = - \sum_{i \in M} \log p_{\theta}(y_i | x_{\text{masked}})$$

where: - M is the set of masked positions. - y_i is the ground-truth token at position i . - $p_{\theta}(y_i | x_{\text{masked}})$ is the predicted probability of token y_i .

Images (Vision Models like SimCLR, MAE)

- **Task**: Maximize similarity between augmented views (SimCLR) or reconstruct masked patches (MAE).
- **Loss Function**: - **Contrastive Loss (SimCLR)**:

$$\mathcal{L} = - \log \frac{\exp(\text{sim}(z_i, z_j)/\tau)}{\sum_{k=1}^N \exp(\text{sim}(z_i, z_k)/\tau)}$$

- **Reconstruction Loss (MAE)**:

$$\mathcal{L} = \frac{1}{n} \sum_{i=1}^n ||x_i - \hat{x}_i||^2$$

Sound (Speech Models like Wav2Vec 2.0)

- **Task**: Learn speech representations by distinguishing correct and incorrect future time steps.
- **Loss Function**: InfoNCE (Contrastive Loss for Audio).

$$\mathcal{L} = - \log \frac{\exp(\text{sim}(q, k^+)/\tau)}{\sum_{k \in K} \exp(\text{sim}(q, k)/\tau)}$$

where: - q is the query (current speech embedding). - k^+ is the correct future speech embedding. - k^- are negative distractor embeddings. - τ is the temperature parameter.

Random Augmentations for Images in SSL

Geometric Transformations

- **Types**: Random cropping, flipping, rotation, affine transformations. - **Purpose**: Alter spatial properties while preserving semantic content.

Photometric Transformations

- **Types**: Color jittering, grayscale conversion, Gaussian blur, solarization. - **Purpose**: Modify pixel values to simulate real-world variations.

Noise-Based Transformations

- **Types**: Gaussian noise, salt-and-pepper noise. - **Purpose**: Introduce random distortions to improve robustness.

Occlusion-Based Transformations

- **Types**: Random erasing, cutout, mixup, cutmix. - **Purpose**: Force the model to learn from partial information.

Summary Table of SSL Techniques

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